Problem Pack: Pythagorean Theorem Applications, Mathematics 8 - Unit 2 Review

Question 17

Problem: 17. A wheelchair ramp rises 1.5 m over a horizontal run of 3.6 m. Find the length of the ramp surface. Round to two decimal places where needed.

Student Work:

1.5^2 + 3.6^2 = c^2 = √15.21 = 3.9

Feedback

Question 18

Problem: 18. A triangular garden has two perpendicular sides measuring 9 m and 12 m. Find the third side, then find the total length of fencing needed to surround the garden.

Student Work:

9^2 + 12^2 = √225 = 15

Feedback

Question 19

Problem: 19. Two boats leave the same marina at the same time. One travels north at 5 km/h and the other travels east at 12 km/h. How far apart are the boats after 2 hours? Show how the situation forms a right triangle.

Student Work:

10^2 + 24^2 = √676 = 26

Feedback

Question 20

Problem: 20. A two-part support is made from two right triangles. In the first triangle, the legs are 7 cm and 24 cm. Its hypotenuse becomes one leg of a second right triangle whose other leg is 60 cm. Find the final length x. Show both uses of the Pythagorean theorem.

Student Work:

7^2 + 24^2 = r^2 = 49 + 576 = √625 = 25 25^2 + 60^2 = √4225 = 65

Feedback